

Backend Development Environment Setup

Windows — .NET / C# / SQL Server
Target stack: .NET 10 / C# 14 • SQL Server 2025 • Git • OS: Windows

Setup Status: Complete

Tool	Version / Detail	Status
1. .NET SDK	10.0.400	Installed — confirmed via <code>dotnet --version</code>
2. Visual Studio Community 2026	.NET desktop dev + ASP.NET/web dev workload	Installed
3. SQL Server 2025 Standard Developer Edition	Standard Developer (not Enterprise)	Installed
4. SQL Server Management Studio	SSMS 22, Core Components only	Installed
5. Git for Windows	2.46.1.windows.1	Already present — confirmed via <code>git --version</code>

Commonly needed later, deliberately not installed as part of this baseline setup:

- Docker Desktop — only needed once a project requires containerization/deployment
- Node.js / Angular CLI — only needed once a project's frontend needs to run against this backend locally

General principle: install the minimum needed to start, add the rest only when an actual feature calls for it.

Download Sources

Tool	Download page
.NET 10 SDK	https://dotnet.microsoft.com/en-us/download/dotnet/10.0
Visual Studio 2026 Community	https://visualstudio.microsoft.com/downloads/
SQL Server 2025 Downloads	https://www.microsoft.com/en-us/sql-server/sql-server-downloads
SQL Server Management Studio (SSMS)	https://learn.microsoft.com/en-us/ssms/install/install
Git for Windows	https://git-scm.com/download/win

Setup Steps — Detail

1. .NET 10 SDK

Download the SDK (not Runtime-only) — the SDK bundles the runtime, CLI, and build tools. Use the x64 installer on standard Intel/AMD Windows machines (Arm64 only needed for ARM-based devices like Surface Pro X or Snapdragon Copilot+ PCs — check via Settings → System → About → System type). Verify post-install with `dotnet --version`.

2. Visual Studio Community 2026

Free for individual/student use (a license condition, not a trial). Recommended workloads: **.NET desktop development** and **ASP.NET and web development** — the desktop workload is required for the Console App project template to appear at all; the web workload covers Web API, EF Core, and NuGet for any web/API project. Skip Azure development, AI Development, and all other offered workloads (Python, C++, Node.js, mobile, game dev, data science) unless a specific project actually needs them. VS 2026 (not VS 2022) is required to fully use .NET 10 / C# 14 tooling in the editor.

3. SQL Server 2025 Standard Developer Edition

Choose Developer edition over Express — Express caps database size at 10GB and strips out features (execution plans, advanced indexing) worth having for real learning/dev work. Within Developer edition, choose **Standard Developer** over Enterprise Developer — SQL Server 2025 split Developer into two variants for the first time, matching either the Standard or Enterprise feature set. Standard Developer is the better default because it matches what most real production environments actually run (Enterprise is expensive and comparatively rare); building against Standard avoids a false sense of what's actually usable later. Basic install type, default instance is sufficient for local development.

4. SQL Server Management Studio (SSMS) 22

Installable directly from the SQL Server installer's own shortcut at the end of setup, or as a separate download — same tool either way. SSMS 22 is the current, only-supported release (Microsoft dropped support for SSMS 21 and earlier once 22 shipped) and is built for full compatibility with SQL Server 2025. Install Core Components only — the five optional add-ons (AI Assistance, Business Intelligence, Hybrid and Migration, Code tools, Database DevOps Preview) are unrelated to day-to-day query/schema/index work and can be added later if a specific need arises. Note: Azure Data Studio was retired in February 2026 — SSMS is the correct primary tool on Windows, not ADS.

5. Git for Windows

Download from git-scm.com if not already present; default install options are fine. Verify with **git --version**. Any reasonably recent version works — core operations (init/clone/commit/push/pull, branching, merge conflict resolution) have been stable across Git versions for years, so there's no need to chase the newest release.

Setup Doubts and Resolutions

Which .NET 10 SDK installer variant?

Question: Should the x64 Windows installer be used?

Answer: Yes — x64 is correct for standard Intel/AMD machines. Only ARM-based Windows devices (Surface Pro X, some Snapdragon Copilot+ PCs) need Arm64. x86 (32-bit) is legacy and not relevant. Confirm via Settings → System → About → System type, and afterward via `dotnet --version`.

VS 2026 Community download not starting in Chrome

Question: Clicking “Free Download” for Visual Studio Community in Chrome did not start the download, even after retrying via the “Click Here to retry” link.

Answer: Opening the same download link in Microsoft Edge resolved it — download started and completed successfully on the first try. This was a browser-specific issue on that machine, not a problem with the installer link itself; worth trying a different browser if a Microsoft download silently fails to start.

Should other VS workloads be selected? (Azure and AI Development, .NET Desktop development)

Question: Beyond the initially suggested workload, should Azure and AI Development or .NET Desktop development also be selected?

Answer: .NET desktop development should be added — Microsoft's own docs confirm the Console App template requires it; ASP.NET and web development alone does not surface it. Azure development and AI Development should be skipped unless a project specifically needs Azure deployment tooling or AI/Semantic Kernel work. All other offered workloads (Python, C++, Node.js, mobile, game dev, data science) should also be skipped by default. Recommended baseline: .NET desktop development + ASP.NET and web development.

Visual Studio Build Tools showing in VS Installer, setup paused

Question: The VS Installer also listed “Visual Studio Build Tools” with its setup paused — should it be removed?

Answer: Not needed for typical local development — safe to cancel if paused/incomplete. VS Build Tools is a separate, IDE-less product for CI/build-server use; the full Community IDE already includes everything it offers once the relevant workloads are installed. One caveat: if some other unrelated program on the machine specifically depends on Build Tools (a common cause is Node.js native module compilation via node-gyp), canceling it would affect that — worth checking before removing if that's a possibility.

SQL Server 2025 Developer: Standard or Enterprise Developer edition?

Question: SQL Server 2025's download page offers both “Standard Developer edition” and “Enterprise Developer edition” under Developer — which one?

Answer: Standard Developer edition, as a general default. SQL Server 2025 split Developer into two variants for the first time; pre-2025, Developer always meant free Enterprise, which could create a mismatch between local dev/test and real Standard-edition production. Standard Developer matches what most real production environments actually run. Enterprise Developer is the right choice only if the target production environment is confirmed to run Enterprise edition.

“Install SSMS” button in SQL Server installer's final popup

Question: After a successful SQL Server install, the final popup offered an “Install SSMS” button — should it be used?

Answer: Yes. It installs the same SSMS as a separate download would — the SQL Server installer just offers a shortcut to it instead of a separate download page. No difference in outcome, just fewer steps.

SSMS 22 installer — correct version?

Question: The “Install SSMS” button led to an SSMS 22 installer — is this the right version?

Answer: Yes. SSMS 22 is the current, only-supported release (Microsoft dropped support for SSMS 21 and earlier once 22 shipped) and is built for full compatibility with SQL Server 2025. One caveat: SSMS 22's official system requirements page lists Windows 11 + Server 2019/2022/2025 only, though Microsoft's own SSMS FAQ elsewhere says “Windows 10 or later” — worth being aware of in case a compatibility warning appears during install on Windows 10.

SSMS 22 install: which optional components/workloads to select?

Question: The SSMS 22 installer's Workloads screen offers five optional add-ons (AI Assistance, Business Intelligence, Hybrid and Migration, Code tools, Database DevOps Preview) alongside the always-included Core Components — which should be selected?

Answer: None, as a general default. Core Components alone covers standard query/schema/index/execution-plan work. AI Assistance, Business Intelligence, Hybrid and Migration, Code tools, and Database DevOps (still in preview) are all specialized add-ons worth installing only if a specific, concrete need for them comes up — same minimal-footprint principle as the VS 2026 workload selection.

General-purpose reference for setting up a .NET / SQL Server backend development environment on Windows. Not tied to any specific project.